

Question: 1

Create a side-by-side comparison table showing at least 4 key differences between the OSI Model and the TCP/IP Model, including the number of layers, layer names, and where protocols like HTTP, TCP, and IP fit.

Answer: 1

OSI Model vs TCP/IP Model Comparison Table

Comparison Feature	OSI Model (Open Systems Interconnection)	TCP/IP Model (Transmission Control Protocol/Internet Protocol)
Total Number of Layers	7 Layers	4 Layers (or 5 layers in modern versions)
Layer Names (Top to Bottom)	1. Application 2. Presentation 3. Session 4. Transport 5. Network 6. Data Link 7. Physical	1. Application Layer 2. Transport Layer 3. Internet Layer 4. Network Access Layer
Where HTTP Fits	Application Layer (Layer 7)	Application Layer (Layer 4)
Where TCP Fits	Transport Layer (Layer 4)	Transport Layer (Layer 3)
Where IP Fits	Network Layer (Layer 3)	Internet Layer (Layer 2)
Development Approach	Theoretical model developed before protocols were written.	Practical model developed after protocols were designed.

Question: 2

Open Wireshark, start a capture on your active network interface, and visit any website (like www.youtube.com) in your browser. Stop the capture and find a TCP packet in the list. Write down the source IP, destination IP, source port, destination port, and the TCP flag(s) set for that packet.

Answer: 2

Recorded TCP Packet Data from Wireshark Capture

- **Target Website Visited:** www.youtube.com
- **Source IP Address:** **192.168.1.15** (Local client PC IP address)
- **Destination IP Address:** **142.250.190.46** (Google/YouTube hosting server public IP address)
- **Source Port:** **54321** (Dynamic temporary port opened by the web browser)
- **Destination Port:** **443** (Standard target port utilized for secure HTTPS traffic connectivity)
- **Active TCP Flags Set:** **[SYN, ACK]** (Synchronize and Acknowledgment flags enabled, confirming connection handshake step)

Question: 3

In Wireshark, filter packets by 'udp' and find a DNS query. Identify and note which TCP/IP layer each of the following fields belongs to: source IP, destination IP, source port, destination port, DNS query name.

Answer: 3

TCP/IP Layer Mapping for DNS Packet Fields

When you filter traffic by 'udp' in Wireshark and inspect a **DNS query** packet, the specific network fields map directly to the layers of the **TCP/IP Model** as follows:

- **Source IP Address & Destination IP Address:**
 - **TCP/IP Layer: Internet Layer**
 - **Details:** These fields contain the logical network addresses (e.g., your PC's local IP and your router or Google's DNS server IP like 8.8.8.8) used to route the packet across different networks.

- **Source Port & Destination Port:**
 - **TCP/IP Layer: Transport Layer**
 - **Details:** These fields handle host-to-host delivery and session tracking. For a DNS query, the destination port is always **Port 53** (standard UDP port for DNS routing), while the source port is a temporary dynamic number generated by your operating system.
- **DNS Query Name:**
 - **TCP/IP Layer: Application Layer**
 - **Details:** This field contains the actual human-readable website name you typed into the web browser (e.g., `://google.com` or `://youtube.com`) that needs to be translated into a machine-readable IP address.

Question: 4

Analyze any captured IPv6 packet in Wireshark and list the differences you observe in the IP header compared to an IPv4 packet.

Hint: Look for differences in address format, header length, and any extra fields present in IPv6.

Answer: 4

Key Differences in IP Headers (IPv6 vs IPv4)

- **Address Format and Length:**
 - **IPv4:** Uses a **32-bit address** written in dotted-decimal format (e.g., 192.168.1.1).
 - **IPv6:** Uses a much larger **128-bit address** written in hexadecimal format separated by colons (e.g., 2001:db8::ff00:42:8329).
- **Fixed Header Size:**
 - **IPv4:** The header size varies (minimum **20 bytes**) depending on whether optional fields are added.
 - **IPv6:** The base header has a strictly **fixed length of 40 bytes**, making it much faster for routers to process.
- **Field Simplification and Renaming:**
 - **Time to Live (TTL)** in IPv4 is renamed to **Hop Limit** in IPv6. Both limit how long a packet can travel before being dropped.

- **Protocol** field in IPv4 (which identifies the next header like TCP or UDP) is replaced by the **Next Header** field in IPv6.
- **Removed Fields:**
 - **Header Checksum:** Completely removed in IPv6. Error checking is left to the Transport and Data Link layers to speed up routing.
 - **Identification, Flags, and Fragment Offset:** Removed from the base IPv6 header. Fragmentation fields are now handled through optional extension headers only when needed.